



OLA Ride Booking Data Analysis Project Report

1. Project Overview

The OLA Ride Booking Data Analysis Project was developed to analyze ride-booking operations and generate business insights using SQL, Excel, and Power BI. The project focuses on understanding booking trends, customer behavior, cancellation patterns, revenue performance, vehicle utilization, and service quality metrics.

The objective of this project is to transform raw booking data into meaningful business intelligence that can support data-driven decision-making.

2. Objectives

The main objectives of this project are:

- Analyze ride booking trends and booking volumes.
- Measure overall revenue generated from ride bookings.
- Identify successful and cancelled booking patterns.
- Evaluate vehicle-wise performance.
- Analyze customer and driver ratings.
- Understand cancellation reasons from both customer and driver perspectives.
- Create an interactive dashboard for business reporting.

3. Tools and Technologies Used

Tool	Purpose
SQL	Data Extraction, Querying and Analysis
Excel	Data Cleaning and Validation
Power BI	Dashboard Creation and Visualization
DAX	KPI and Measure Calculations

4. Dataset Information

The dataset contains ride booking information from a ride-sharing platform.

Dataset Features

- Booking ID
- Booking Status
- Customer ID
- Vehicle Type
- Booking Value
- Ride Distance
- Payment Method
- Driver Rating
- Customer Rating
- Cancellation Reasons
- Date Information

Dataset Size

- Records: 20,407+
 - Columns: 19
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5. Data Preparation

The dataset was imported and validated before analysis.

Data Cleaning Activities

- Checked for missing values.
 - Verified data types.
 - Standardized column formats.
 - Validated booking status categories.
 - Cleaned cancellation reason fields.
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6. SQL Analysis

SQL was used to answer various business questions.

Key SQL Queries

1. Total Successful Bookings
2. Average Ride Distance by Vehicle Type

3. Revenue Analysis
4. Customer Rating Analysis
5. Driver Rating Analysis
6. Cancellation Analysis
7. Vehicle Performance Analysis
8. Booking Status Breakdown

SQL Views were created to improve query management and reporting efficiency.

7. Power BI Dashboard Development

The dashboard was designed with five analytical sections.

Dashboard Pages

1. Overall Dashboard

Key Metrics:

- Total Bookings
- Total Revenue
- Ride Volume Trend
- Booking Status Distribution

Purpose: Provides a high-level overview of business performance.

2. Vehicle Type Dashboard

Key Metrics:

- Vehicle-wise Revenue
- Successful Booking Revenue
- Average Distance Travelled
- Total Distance Travelled

Purpose: Measures performance across different vehicle categories.

3. Revenue Dashboard

Key Metrics:

- Revenue by Payment Method

- Date-wise Ride Distance
- Top Customers

Purpose: Analyzes revenue generation and customer contribution.

4. Cancellation Dashboard

Key Metrics:

- Total Bookings
- Successful Bookings
- Cancelled Bookings
- Cancellation Rate

Additional Analysis:

- Customer Cancellation Reasons
- Driver Cancellation Reasons

Purpose: Identifies operational issues affecting booking completion.

5. Ratings Dashboard

Key Metrics:

- Driver Ratings
- Customer Ratings

Purpose: Measures service quality and customer satisfaction.

8. DAX Measures Created

Total Bookings

Calculates total number of bookings.

Successful Bookings

Calculates completed rides.

Cancelled Bookings

Calculates total cancelled bookings.

Cancellation Rate

Formula:

Cancellation Rate = (Cancelled Bookings / Total Bookings) × 100

Revenue Metrics

Used to calculate booking value and revenue trends.

9. Key Insights

Booking Performance

- More than 20,000 bookings were analyzed.
- Successful bookings contributed the majority of completed rides.

Revenue Insights

- Total revenue exceeded 7 million.
- Cash and UPI were the dominant payment methods.

Cancellation Analysis

- Cancellation rate was approximately 28%.
- Driver-related operational issues were a major reason for ride cancellations.
- Customer-driven cancellations were also observed due to changes in travel plans and service-related concerns.

Vehicle Performance

- Prime Sedan generated the highest booking value.
- Auto category showed the lowest average ride distance.

Ratings Analysis

- Driver ratings remained close to 4.0.
 - Customer ratings were also consistently around 4.0, indicating stable service quality.
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10. Business Impact

This analysis can help ride-sharing companies:

- Reduce ride cancellations.
- Improve customer satisfaction.

- Optimize vehicle allocation.
 - Increase operational efficiency.
 - Improve driver performance management.
 - Enhance revenue generation strategies.
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11. Conclusion

The OLA Ride Booking Data Analysis Project demonstrates the complete analytics workflow from data preparation to business intelligence reporting. By leveraging SQL, Excel, Power BI, and DAX, meaningful insights were generated regarding booking behavior, revenue performance, vehicle utilization, customer satisfaction, and cancellation trends.

This project showcases practical Data Analytics skills and provides a strong portfolio project for aspiring Data Analysts.